

Knowledge Units

Definition

A Knowledge Unit (KU) is a thematic grouping that encompasses multiple, related KU outcomes and learning topics.

A Knowledge Unit (KU) outcome is a specific assessment of a concept associated with a particular KU.

Model KU Structure

Name: The name used to identify a knowledge unit. The name is followed by a three letter key in parenthesis.

The key is for indexing in data structures.

Description: A short narrative description of the scope and contents of the knowledge unit. The intent of this knowledge unit is to provide students with a [basic/intermediate/advanced] awareness of [details].

Outcomes: A description of student based outcomes associated to the knowledge unit.

Students will be able to [outcome #1].

Students will be able to [outcome #2].

KU Topics: A list of elements in the KU. These topics should be listed in an appropriate hierarchy of detail. The format of the topics element should appear as follows:

High level name 1 – description of the high level name

Sub level name 1 – description of first sub level element

Sub level name 2 – description of second sub level element

High level name 2 – description of the high level name

...

High level name N – description of the high level name

CSF. Cybersecurity Fundamentals

This KU covers the current cybersecurity threat landscape, including the various motivations, objectives, and threats. This knowledge reinforces adversarial thinking and the need for cybersecurity. Students must also be able to describe cybersecurity goals, how risk management informs the approach to cybersecurity, and the principles and best practices of cybersecurity.

KU Learning Outcomes

Students will be able to:

1. Define the fundamental principles of cybersecurity.
2. Describe the goals of cybersecurity.
3. Describe the potential objectives of cyber attackers and their motivations.
4. Describe various types of cyber attacks.
5. Describe various common vulnerability types and evaluate their severity.
6. Describe different types of malware.
7. Describe common mechanisms used to protect against cyber attacks.
8. Given a specific scenario, identify potential protection mechanisms and design principles.
9. Describe the components of access control and how they are implemented.
10. Discuss current events in cybersecurity

Topics:

1. Cybersecurity principles
2. Adversarial thinking
3. Threat actors
4. Attack methods, including social engineering
5. Vulnerabilities
6. Vulnerability taxonomies, databases, and scores
7. Malware
8. Cybersecurity/McCumber Cube (Confidentiality, Integrity, and Availability (CIA) Triad, Information States, Security Controls)
9. Authentication, authorization, and accounting (AAA) and access control
10. Cybersecurity best practices
11. Cyber risk management
12. Current events in cybersecurity

Notes:

In undergraduate programs, this KU will likely be covered by an introductory cybersecurity course. For graduate programs, this KU may be covered in the introductory portion of a more advanced cyber course. The intent is to match with CAE-CD.

ISC. IT Systems Components

This KU covers a basic understanding of the components of an information technology system and their roles in system operation. It provides a high-level introduction or familiarization with the topics rather than a deep dive into specifics

KU Learning Outcomes

Students will be able to:

1. Describe the various hardware components of modern computing environments and their individual functions.
2. Characterize the networks that interconnect computing components.
3. Describe the various IT activities and roles involved in managing computing systems.
4. Understand system and software architectures.
5. Describe how software is managed within an organization.
6. Understand basic security issues in modern computing environments.

Topics:

1. Endpoint protection (workstations, personal computers, servers, appliances, mobile devices, peripherals)
2. Storage Devices (external storage drives, filesystems, shared file servers, databases)
3. System Architectures (enterprise, cloud, SaaS/PaaS/IaaS, etc.) and enabling technologies (containers, virtualization)
4. Managed Services Alternative environments (SCADA, real-time systems, critical infrastructures)
5. Configuration Management
6. System and application software updates patching
7. Networks (Internet, LANs/WANs, wireless, Internet of Things)
8. Network mapping (identification/enumeration of networked components)
9. Cyber-physical systems (SCADA, real-time systems, critical infrastructure)
10. Network security components (firewalls, VPNs, monitors, intrusion detection systems)
11. IT organizational structure

Notes:

This KU will likely be covered in undergraduate programs by an introductory networking and/or general computing course. Examples of hands-on activities include configuring network testbeds (either real or emulated). This KU may be covered by prerequisite or prior undergraduate coursework for graduate programs. The intent is to match with CAE-CD.

BSP. Basic Scripting and Programming

This KU covers foundational skills in scripting and programming, focusing on enabling students to write simple scripts to automate tasks and perform basic programming operations.

KU Learning Outcomes

Students will be able to:

1. Write simple scripts to automate system administration tasks.
2. Write programs in a high-level language using basic programming constructs and concepts.
3. Implement basic security practices in scripting, including bounds checking and input validation.
4. Demonstrate conceptual understanding and practical use of regular expressions.
5. Demonstrate the use of basic data structures and algorithms.

Topics:

1. Fundamental programming constructs
 - a. Variables and types
 - b. Operators (assignment, arithmetic, logic, etc.)
 - c. Arrays, strings, and structures
 - d. Control structures (conditional execution, selection, loops, branching)
 - e. Functions/procedures/subroutines
2. Boolean logic
3. Basic data structures
4. Algorithms
5. Console and file I/O
6. Libraries
7. Program safety (bounds checking, input/parameter validation, type checking)
8. Scripting platforms in various operating systems
9. Sequential and parallel execution
10. Testing and debugging techniques

Notes:

The goal of this KU is to be replaced by the updated CAE-CD Basic Scripting and Programming KU.

This KU is best covered by an introduction to programming course. Python programming is recommended but other languages are acceptable.

The Basic Scripting and Programming KU from the CAE-CD designation or the Low-Level Programming Languages KU from the CAE-CO designation will cover this KU if appropriately supplemented.

MAF. Math Fundamentals

This KU covers the fundamental mathematics necessary to understand and apply AI algorithms and models and analyze AI algorithm outputs. It focuses on essential aspects of algebra, probability, and applied statistics.

KU Learning Outcomes

Students will be able to:

1. Apply basic algebraic operations and solve linear equations, inequalities, and systems of equations.
2. Analyze and graph various types of functions, including linear, quadratic, polynomial, rational, exponential, and logarithmic functions.
3. Perform operations with functions, including composition and finding inverses.
4. Solve quadratic, higher-degree polynomial, exponential, and logarithmic equations.
5. Use algebraic models to represent and solve real-world problems, including those involving sequences and series.
6. Understand and differentiate between descriptive and inferential statistics, including the use of statistical measures and distributions.
7. Organize and summarize data using tables, charts, and graphs, and interpret measures of central tendency and variability.
8. Understand and apply fundamental probability principles

Topics:

1. Equations and the concept of function: linear, quadratic, higher-degree polynomial, exponential, logarithmic, and rational functions; complex numbers.
2. Probability measures and distributions
3. Descriptive Statistics (mean, variance, median, etc.)

Notes:

This KU is likely covered by multiple math courses in an undergraduate program. In a graduate program, it may be covered by prerequisite or prior undergraduate coursework.

AIG. AI Governance, Laws, and Ethics

This KU covers the governance, legal, and ethical considerations surrounding AI systems, including the frameworks and principles that guide the responsible development and deployment of AI technologies.

KU Learning Outcomes

Students will be able to:

1. Analyze AI Governance related to the US and International legal and regulatory frameworks, organizational policies, and oversight mechanisms.
2. Describe the societal responsibility of AI developers, operators, and users to act legally and ethically at all times.
3. Describe AI ethical considerations (Fairness(bias)), Transparency, Accountability, and Non-discrimination).
4. Assess the impact of laws and regulations on AI development and security.

Topics:

1. AI Governance
 - a. Legal and Regulatory Frameworks (domestic and international)
 - b. Standards and standard-setting bodies for AI development and deployment.
 - c. Internal organizational policies guiding legal and ethical AI use.
 - d. Oversight and Compliance mechanisms for monitoring and enforcing AI regulations and policies.
 - e. Stakeholder Engagement (Boards of Directors, Leadership, Customers, Public)
2. AI Laws and Regulations, including assessment of practical examples/case studies
 - a. Data Protection and Privacy - GDPR, CCPA, HIPAA, EU AI Act, organizational retention policies
 - b. Accountability and Liability
 - c. Non-Discrimination
 - d. International Alignment
3. Develop policies guiding data privacy and protection related to AI.
4. Ethical Considerations (transparency, fairness, accountability, beneficence, privacy, trust/trustworthiness) and frameworks

Notes:

This KU may overlap with similar KUs from the CAE-CD program or study or the CAE-CO program of study.

AIF. AI Fundamentals

This KU covers the basic concepts of Artificial Intelligence (AI). It includes basic security principles surrounding artificial intelligence and its applications.

KU Learning Outcomes

Students will be able to:

1. Describe the principles and concepts of AI, including terminology, components, tools, various AI technologies, and their evolution.
2. Explain classical artificial intelligence algorithms, such as search and optimization algorithms.
3. Explore and assess how AI technologies can transform cybersecurity practices, including enhancing security protocols and predicting security risks (e.g., threats and vulnerabilities).
4. Evaluate the suitability of an AI tool for a specific task, particularly within the context of cybersecurity.
5. Discuss applications of Artificial Intelligence techniques in cybersecurity and related domains.
6. Describe the ethical considerations and potential risks associated with AI, focusing on bias, privacy, safety, and security throughout the lifecycle.
7. Identify solutions to business problems by utilizing intelligent systems and machine learning.

Topics:

1. Overview and evolution of the broad field of Artificial Intelligence
2. Classes of AI tools and techniques (classical AI, expert systems, machine learning, neural networks, generative AI)
3. Problem discovery and formulation
4. Knowledge representation and reasoning
5. AI tools and frameworks
6. Classic AI algorithms (search, optimization, rule-based systems, decision trees)
7. Intelligent agents
8. Safety and security principles of AI
9. AI risk and ethics

Notes:

Topics in this KU are intended to be covered broadly and thus at a relatively high level of depth. Later KUs build on this one to provide selective depth on respective topic areas.

MLF. Machine Learning Fundamentals

This KU covers fundamental concepts of Machine Learning (ML) and their applications in cybersecurity. It includes basic principles, techniques, and tools used in Machine Learning, with an emphasis on their applications to cybersecurity domains.

KU Learning Outcomes

Students will be able to:

1. Define machine learning and discuss frequently used machine learning methods.
2. Describe various machine learning algorithms.
3. Compare and contrast various ML methods.
4. Identify the classes of problems for which machine learning is best suited.
5. Evaluate the performance and effectiveness of different Machine Learning methods and tools.
6. Identify various data types and formats used by AI systems.
7. Apply Machine Learning methods effectively for cybersecurity applications.
8. Analyze security case studies and threats using Machine Learning techniques.

Topics:

1. Introduction to ML tools and frameworks
 - a. Overview of commonly used ML tools and frameworks
 - b. Basic model training, testing, and validation
2. Common ML algorithms
 - a. Introduction to decision trees, support vector machines, and K-nearest neighbors
 - b. Basic applications and understanding of how these algorithms function
3. Learning paradigms
 - a. Introduction to supervised learning: neural networks, decision trees, and regression
 - b. Introduction to unsupervised learning: clustering and dimensionality reduction
 - c. Overview of semi-supervised and reinforcement learning
4. Data Types and Pre-Processing
 - a. Common data formats and standards: CSV, JSON, binary
 - b. Basic data pre-processing techniques: deduplication, outlier management, normalization, and feature extraction
5. Introduction to Neural Networks and Deep Learning:
 - a. Basic neural network architectures
 - b. Introduction to deep learning concepts: convolutional and recurrent neural networks, long short-term memory (LSTM)
 - c. Overview of parameter tuning, overfitting, and mitigation strategies
6. ML Applications in Cybersecurity
 - a. Application of ML methods in cybersecurity: anomaly detection, attack detection, network traffic analysis
 - b. Case studies in cybersecurity using ML techniques
7. Introduction to Specialized ML Applications:
 - a. Overview of Natural Language Processing (NLP)
 - b. Basic concepts of generative AI: large language models, prompt engineering, and fine-tuning

Notes:

AIS. AI for Security Assessments

This KU covers the principles and practices of conducting security assessments and evaluations leveraging AI. It includes methodologies for assessing organizations, evaluating vulnerabilities, and implementing measures to mitigate identified risks.

KU Learning Outcomes

Students will be able to:

1. Conduct security evaluations using AI enhancements to standard tools and techniques
2. Produce comprehensive reports with actionable recommendations
3. Leverage AI to assess the impact of identified vulnerabilities and design and implement mitigation strategies
4. Perform threat modeling and develop mitigation strategies
 - a. Map attack surfaces and develop comprehensive threat models

Topics:

1. Security Assessment Methodologies
 - a. Tools and frameworks
 - b. Risk and Vulnerability Assessment
2. Assessing and conducting monitoring and detection leveraging AI
3. Threat modeling, monitoring, and threat detection
 - a. Taxonomy of attacks and defense

Notes:

DFA. Defensive Applications of AI

This KU covers the application of AI in strengthening defensive cybersecurity measures. Students will learn how AI-driven tools and techniques can improve the detection, analysis, and response to cyber threats.

KU Learning Outcomes

Students will be able to:

1. Implement AI techniques to enhance security posture and automate cybersecurity processes.
2. Develop and apply AI and machine learning models to identify cyber risks (e.g., vulnerabilities and threats) and their mitigation.
3. Understand frameworks like MITRE ATLAS for AI-driven cyber threat intelligence.
4. Understand Explainable AI-driven cybersecurity solutions.

Topics:

1. Internal cybersecurity data sources, e.g., network flows, malware, log files, etc.
2. AI Techniques for Cybersecurity
 - a. Machine and Deep learning for cybersecurity
 - b. Reinforcement learning for cybersecurity
 - c. Predictive modeling
 - i. Supervised and unsupervised learning models in threat detection
 - ii. Anomaly and intrusion detection techniques
 - iii. Training and testing ML models with cybersecurity datasets
3. Cyber Threat Intelligence and Analysis Frameworks
 - a. MITRE ATLAS framework
 - b. NIST Risk Management Framework
 - c. Natural Language Processing (NLP) for Cyber Threat Intelligence
 - d. Malware analysis and detection (e.g., Reverse Engineering)
4. Defensive AI Applications
 - a. AI-based Network Intrusion Detection Systems (NIDS)/Host Intrusion Detection System(HIDS)
 - b. AI-based vulnerability defense and repair mechanisms
 - c. AI for blue team operations/ Incident Response
 - d. AI for digital forensics
5. Explainable AI(XAI)-driven cybersecurity solutions
 - a. Understanding the limitations of AI-driven solutions
 - b. Post-hoc evaluation (SHAP, LIME)
 - c. Interpretable models and representations

Notes:

OFA. Offensive Applications of AI

This KU covers the application of Artificial Intelligence (AI) in conducting offensive cybersecurity operations. It includes strategies for leveraging AI in penetration testing, exploiting vulnerabilities, and automating cyber attacks.

KU Learning Outcomes

Students will be able to:

1. Identify and explain adversarial attacks on machine learning models
2. Identify opportunities for AI to enable offensive cybersecurity operations.
3. Supplement current offensive Tactics, Techniques, and Procedures (TTPs) with AI.
4. Evaluate the effectiveness of implementing AI for conducting cyber attacks.
5. Discuss the ethical and legal implications of AI-enabled cyber offensive operations.
6. Correlate offensive AI techniques to existing cybersecurity frameworks

Topics:

1. Basic of adversarial learning
2. External cybersecurity data sources, e.g., open source intelligence (OSINT), Dark Web, internet of things search engines, cybersecurity data repositories, etc.
3. AI techniques for offensive cybersecurity
 - a. Vulnerability discovery (e.g., scanning, code analysis, reverse engineering)
 - b. Penetration testing and exploitation (e.g., automated exploit generation, CVE-to-exploit)
 - c. Reconnaissance (e.g., active/passive scanning, OSINT collection/aggregation)
4. Use of AI for attack model frameworks (e.g., cyber kill chain, ATT&CK)
5. AI-enhanced red-teaming TTPs
6. Ethical and legal considerations for AI-enabled cybersecurity operations
7. White-box vs gray-box vs black-box attacks
8. Formulation and mapping objectives to AI algorithms and TTPs.

Notes:

This will be added as an optional KU for the CAE-CO.